Spatially Explicit Rock–TBM Interaction Graph Sequences for Response-Consistent Interpretation of TBM Excavation Processes

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Abstract

Existing representations of rock–TBM interaction during excavation treat monitoring responses as temporal sequences, which preserve temporal dependence but lack the spatial explicitness needed to resolve where interactions occur, which machine components are implicated, and how interaction structure evolves along chainage. This study proposes a response-supervised, geometry-constrained graph sequence framework for spatially explicit representation and interpretable learning of rock–TBM interaction. The framework introduces three key innovations: (i) a heterogeneous graph-sequence representation that formalizes TSP-derived rock voxels and parameterized TBM surface nodes as spatial entities with explicit candidate interaction relations; (ii) a geometry-constrained edge construction strategy based on spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints, which restricts learning to spatially plausible rock–machine relations rather than unconstrained connectivity; and (iii) a graph-to-surface interpretation route that projects learned edge relevance onto TBM surface and chainage spaces to produce response-consistent hotspot maps and interaction-evolution views. Rather than relying on sparse jamming labels or manually calibrated contact forces, the framework uses continuously recorded monitoring responses to supervise the relative importance of geometry-screened candidate relations, enabling interpretable spatial learning under incomplete observability. The evaluation compares the framework against monitoring-only sequence models, TSP-augmented sequence models, tabular baselines, and structural graph ablations. Spatial consistency indicators—including Moran's I, component-level coefficient of variation, and attention–geology correlation—suggest that geometry-constrained graph construction produces spatially structured and geologically aligned interaction patterns that degrade when geometric constraints are removed. By unifying geological sensing, machine geometry, monitoring response, and spatial interpretation within a constrained graph sequence, the framework offers a spatially explicit approach to excavation interaction analysis that complements existing temporal prediction methods.

Keywords: spatial representation; heterogeneous graph sequence; rock–machine interaction; geometry-constrained edge construction; response supervision; interpretable geospatial learning; TBM excavation; spatial interpretation
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1 Introduction

Tunnel boring machines (TBMs) are widely used in long and deep tunnel construction, where excavation responses emerge from the coupled effects of geological conditions, machine geometry,

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and operational process. Under fractured, water-rich, high-stress, or locally deformable ground conditions, the interaction state between surrounding rock and the TBM may change rapidly and manifest through abnormal thrust, torque, penetration, advance rate, or shield-related responses. These monitoring variables are informative external expressions of rock–TBM interaction, but monitoring curves alone provide limited evidence about where the interaction occurs, which machine components are implicated, and how local interaction relations evolve along chainage.

Recent developments in tunnel seismic prediction (TSP), advance geological prediction, digital tunnelling, and TBM monitoring systems have substantially increased the availability of heterogeneous excavation data [15]. TSP provides a spatially distributed description of the geological field ahead of or around excavation, whereas monitoring systems provide temporally continuous responses referenced to excavation progress. The challenge is therefore not only data acquisition, but also data organization: heterogeneous sources with different spatial support, temporal resolution, and physical meaning must be integrated into a representation that preserves excavation evolution together with local spatial structure.

This challenge can be viewed as a geographic information representation problem. Three GIScience concerns are central. First, *spatial entity formalisation*: TSP voxels, TBM surface components, and monitoring locations are heterogeneous spatial objects with distinct spatial supports (point, surface, volume) and semantic roles, and must be abstracted into a unified entity model that preserves their geometric and topological properties [7, 9, 13]. Second, *constrained relation construction*: candidate rock–machine relations cannot be learned freely from black-box correlation but must respect geometric compatibility, excavation state, and active-zone constraints—an instance of the broader GIScience principle that spatial relations should be constructed with domain-plausible topology rather than unconstrained connectivity [6, 19, 36]. Third, *interpretable spatial learning*: the learned representation must be projectable back to interpretable spatial views (surface hotspots, chainage evolution) to support spatial reasoning, not merely optimise a prediction loss [14, 17]. In this light, the problem extends beyond conventional engineering prediction and can be framed as a GIScience problem—one that benefits from explicit spatial entity modelling, constrained relation construction, and interpretable spatial learning rather than merely feature concatenation or temporal sequence extension.

Existing TBM prediction and warning studies have shown that tabular models and sequence models can be effective for forecasting penetration, thrust, torque, or jamming-related responses [2, 11, 29, 31]. However, most of these methods represent excavation either as a feature table or as an ordered monitoring sequence. Sequence models are effective for learning temporal dependence in monitoring records [12, 23], but they usually treat excavation as an ordered signal sequence rather than as a spatially structured interaction process. As a consequence, they have limited ability to preserve rock location, machine-component structure, and local rock–machine candidate relations within a unified representation. This limitation matters when the objective is not only response prediction, but also spatially consistent interpretation of interaction dynamics.

Graph learning offers a natural alternative because it is well suited to irregular objects and relations. Graph-based models have shown strong capacity for encoding object interaction, state propagation, and structured spatial systems in domains such as physics simulation [20, 21], urban flow prediction [8, 33], and spatio-temporal forecasting [16, 28]. Heterogeneous graph networks further enable the representation of multi-type entities and relations [22, 26, 32]. Yet TBM excavation is not a generic graph-learning problem in which edges can be learned freely from black-box correlation. Candidate rock–TBM relations must remain consistent with machine geometry, excavation state, and local active zones. In this setting, graph learning is most meaningful when it operates on geometrically plausible candidate relations rather than on unconstrained connectivity.

Motivated by this gap, this study proposes a response-supervised, geometry-constrained graph sequence framework for spatially explicit rock–TBM interaction representation and interpretation. TSP-derived geological attributes are organized as rock voxel nodes, while the cutterhead and shield are discretized as parameterized TBM surface nodes. Candidate rock–machine edges are screened through spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints. Instead of interpreting edge weights as direct physical contact forces or relying on sparse jamming labels, the framework uses continuously recorded monitoring responses to supervise the relative importance of geometry-screened candidate relations. The learned relevance is then projected back to TBM surface space and chainage space to generate response-consistent hotspot and evolution views.

The study addresses three questions: how to organize TSP rock attributes, TBM surface structure, and rock–machine candidate relations into a dynamic graph sequence that preserves spatial explicitness and relation plausibility; how to learn the relative importance of candidate relations from continuous monitoring responses without sparse event labels, enabling interpretable spatial learning under incomplete observability; and how to map learned interaction relevance back to interpretable spatial views that support spatial reasoning beyond prediction. The main contributions are fourfold.

1. *A spatially explicit heterogeneous graph-sequence representation for rock–machine interaction.* Unlike feature-table or pure-sequence representations that discard spatial structure, the proposed representation formalises TSP voxels, TBM surface components, and their candidate relations as typed spatial entities with explicit geometric and topological properties, preserving spatial explicitness throughout the learning pipeline.

2. *A geometry-constrained relation construction strategy that embeds domain plausibility into graph topology.* Candidate rock–machine edges are screened by spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints. This ensures that the learned relations remain geometrically plausible, addressing the GIScience principle that spatial relations should be constructed with domain-plausible topology rather than unconstrained connectivity.

3. *A response-supervised relation learning paradigm that operates under incomplete observability.* Without requiring contact-force labels or sparse jamming events, the framework uses continuously recorded monitoring responses to supervise the relative importance of candidate relations, enabling interpretable spatial learning in settings where ground-truth interaction labels are unavailable.

4. *A graph-to-surface interpretation route that supports spatial reasoning beyond prediction.* Learned interaction relevance is projected back to TBM surface hotspots and chainage-evolution views, accompanied by spatial consistency indicators (Moran's I, component CV, attention–geology correlation) that quantify the spatial plausibility of the learned representation.

2 Methodology

This section formalizes the proposed framework as a geometry-constrained heterogeneous graph sequence supervised by continuously recorded TBM monitoring response. The method represents excavation as an evolving system of spatial entities—rock units and TBM surface components—and their candidate interaction relations, rather than as a monitoring-only signal sequence. By formalizing spatial entities, constraining candidate relations by geometric compatibility, and projecting learned relevance back to interpretable spatial views, the framework addresses the geographic-information representation problem identified in the Introduction: how to organize heterogeneous spatial entities into a representation that preserves spatial explicitness, relation plausibility, and interpretability.

2.1 Framework overview

Let excavation advance be discretized into chainage-indexed steps $t = 1, 2, \ldots, T$. For each step, the framework constructs a graph snapshot G_t and then models the ordered graph sequence $\{G_t\}_{t=1}^{T}$. The workflow contains three stages: geometry-constrained graph-sequence construction, response-supervised interaction learning, and graph-to-surface spatial interpretation (Figure 1). Given the most recent K excavation steps, the model uses graph snapshots together with synchronized TBM monitoring vectors to predict a future response at step $t + h$:

$$\hat{r}_{t+h} = f_\Theta \left(G_{t-K+1:t}, u_{t-K+1:t} \right),$$

where u_t denotes the monitoring vector at step t, r_{t+h} denotes the target future response, K is the historical window length, and h is the forecasting horizon. In this study, the learned rock–machine edge attention is interpreted as response-consistent interaction relevance under the forecasting task, not as true contact force, contact pressure, jamming probability, or any direct engineering risk quantity.

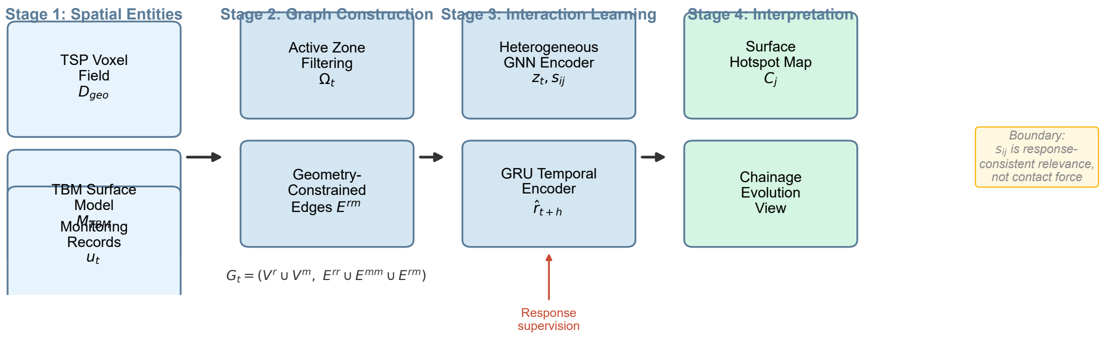

Figure 1: Framework pipeline (schematic). Node counts and edge counts are downsampled for visual clarity; the actual graph uses 1352 TBM surface nodes and approximately 3500 active-zone rock nodes per step (see Implementation details). Stage 1 constructs spatial entities from the TSP voxel field, TBM surface geometry, and monitoring records. Stage 2 builds geometry-constrained graph snapshots through active-zone filtering and candidate rock–machine edge construction. Stage 3 performs response-supervised graph-sequence learning using a heterogeneous GNN encoder and GRU temporal encoder. Stage 4 projects learned relevance back to TBM surface and chainage spaces to generate hotspot maps and evolution views.

2.2 Excavation-step definition and data alignment

Raw TBM monitoring records are usually sampled at a much higher frequency than the graph sequence should use. The method therefore first aggregates monitoring data by chainage or ring-based excavation progress to form a unified excavation-step index t. A practical initial aggregation granularity is 0.05 m or 0.1 m, subject to data density and desired spatial resolution. Each excavation step is associated with four objects: current face or shield position, graph snapshot G_t, synchronized monitoring vector u_t, and future response label r_{t+h}.

The monitoring input vector can include

$$u_t = [\text{AdvanceRate}_t,\ \text{RPM}_t,\ \text{Torque}_t,\ \text{Thrust}_t,\ \text{Penetration}_t,\ \text{ShieldPressure}_t],$$

while the prediction target can be defined as a multivariate future response vector

$$r_{t+h} = [\text{AdvanceRate}_{t+h},\ \text{Torque}_{t+h},\ \text{Thrust}_{t+h},\ \text{Penetration}_{t+h},\ \text{ShieldPressure}_{t+h}].$$

Because TSP coordinates and TBM shield mileage are usually recorded in different local reference systems, the method also requires a local offset mapping. A simple alignment is to define

$$x_{\text{local}}(t) = \text{ShieldMileage}(t) - \text{ShieldMileage}_{\text{start}},$$

and then map $x_{\text{local}}(t)$ into the TSP coordinate range used for voxelized geological representation.

2.3 Geological and machine inputs

The geological input is a TSP-derived voxel field $D_{\text{geo}} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ denotes the spatial coordinate of rock voxel i, and g_i denotes its geological attribute vector. Depending on the available inversion and interpretation products, g_i may include seismic velocity, elastic indicators, ratio features, or other derived geological descriptors.

The TBM input is represented as a parameterized surface model $M_{\text{TBM}} = \{p_j(0), n_j, \rho_j\}$, where $p_j(0)$ is the initial coordinate of TBM surface node j, n_j is the local surface normal, and ρ_j is a machine component label such as cutterhead or shield. The excavation process is further described by advance direction d, step length Δx, step-dependent active zone Ω_t, interaction distance threshold τ, and normal-compatibility threshold $\eta_{\min}$. When a detailed three-dimensional TBM model is unavailable, the surface can be approximated by a parameterized geometry in which the cutterhead is represented as a disc and the shield as cylindrical surface components. In the current formulation, the surface is partitioned into cutterhead, front shield, middle shield, and tail shield, and nodes are sampled at fixed radial, circumferential, and axial resolutions.

2.4 Graph-sequence construction

At each excavation step t, a graph snapshot is defined as

$$G_t = \left(V_t^r \cup V_t^m,\ E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}\right),$$

where V_t^r and V_t^m are rock and TBM node sets, respectively, E_t^{rr} denotes rock–rock adjacency, E_t^{mm} denotes TBM-surface adjacency, and E_t^{rm} denotes candidate rock–machine interaction edges.

Rock node features are defined as

$$x_i^r(t) = [c_i, g_i, s_i(t)],$$

where $s_i(t)$ records whether a voxel remains active and eligible for interaction at step t. TBM node features are defined as

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j].$$

With constant advance step length, TBM surface nodes are updated along the advance direction by

$$p_j(t+1) = p_j(t) + \Delta x\, d.$$

For intra-rock connectivity, E_t^{rr} is constructed from voxel adjacency, for example a 26-neighbourhood relation. For TBM-surface connectivity, E_t^{mm} is constructed from local surface neighbourhood relations inherited from the discretized machine geometry. The key design choice concerns the candidate interaction edges E_t^{rm}, which are not fully connected but screened by geometric plausibility. This constrained construction contrasts with unconstrained graph learning [27, 34] and reflects the domain requirement that candidate rock–machine relations must be physically plausible.

2.5 Geometry-constrained candidate interaction edges

A rock voxel i and TBM surface node j are linked at step t only if the candidate relation satisfies a geometry-constrained rule:

$$e_{ij}^{rm}(t) = 1$$

if

$$\begin{cases} d_{ij}(t) \le \tau, \\ \kappa_{ij}(t) \ge \eta_{\min}, \\ c_i \in \Omega_t, \\ s_i(t) = 1, \end{cases}$$

and the edge is absent otherwise. Here,

$$d_{ij}(t) = \|c_i - p_j(t)\|$$

is the rock-to-surface distance, and

$$\kappa_{ij}(t) = \max\left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon}\right)$$

measures signed-normal compatibility between the TBM surface orientation and the rock-voxel direction. The active-zone condition restricts candidate relations to the geological region currently relevant to excavation, while the state variable $s_i(t)$ prevents already excavated or inactive voxels from repeatedly contributing to later interaction snapshots.

This design intentionally limits graph learning to a screened candidate set. The edge relevance learned later in the model is therefore interpreted as the relative importance of geometrically plausible candidate interactions, rather than as a direct estimate of physical contact force or stress.

2.6 Edge features and geometric prior

Each candidate rock–machine edge is further associated with an edge-feature vector

$$a_{ij}^{rm} = \left[d_{ij}/\tau,\ \kappa_{ij},\ (c_i - p_j)/\tau,\ g_i,\ \mathrm{onehot}(\rho_j)\right],$$

which combines three kinds of information: geometric compatibility, local geological context, and machine-component semantics. In addition, a simple geometric prior is defined as

$$\pi_{ij}^{rm} = \exp(-d_{ij}/\tau)\,\kappa_{ij}.$$

This prior increases when the rock voxel is closer to the TBM surface and more consistent with the local outward normal. It is used only as geometric guidance for learning and should not be interpreted as contact force, contact pressure, or a direct risk indicator. The full graph-sequence construction procedure is summarised in Algorithm 1.

2.7 Response-supervised interaction learning

After graph construction, each snapshot is encoded by a graph encoder to produce an interaction-aware latent representation. Message passing integrates geological attributes, machine-surface context, and candidate interaction structure across the heterogeneous graph. The resulting stepwise graph embeddings are then passed to a temporal encoder, which models graph-state evolution along chainage and links the sequence of graph states to continuously recorded TBM monitoring response.

For each graph snapshot, the graph encoder computes raw (pre-softmax) edge scores $s_{ij}^{rm}(t)$ for each candidate rock–machine edge, which are then normalised over all source rock nodes incident to each destination TBM node j to obtain attention weights $\alpha_{ij}^{rm}(t) = \exp(s_{ij}^{rm}(t)) / \sum_{i'} \exp(s_{i'j}^{rm}(t))$ used in message passing [24, 25]. The encoder also produces a graph-level representation z_t. The graph-level representation is concatenated with the synchronized monitoring vector u_t and passed to a gated recurrent unit (GRU) [4], which models graph-sequence evolution across consecutive excavation steps.

The supervision signal is provided by monitoring response variables recorded during excavation, rather than by sparse binary jamming labels or manually assigned edge weights. Under this design, the model learns which screened candidate rock–machine relations are most relevant for explaining or anticipating response variation. The resulting raw scores $s_{ij}^{rm}(t)$ are therefore response-consistent relevance indicators under geometric screening. The training procedure is summarised in Algorithm 2.

2.8 Graph-to-surface spatial interpretation

To recover spatial interpretability, the learned relevance associated with candidate rock–machine relations is projected back onto two explicit spaces.

Notation. We distinguish three quantities in the interpretation pipeline:

- *Raw edge score* $s_{ij}^{rm}(t)$: the pre-softmax attention score computed by the graph encoder for candidate edge (i, j) at step t. This quantity preserves absolute magnitude differences across edges and is used for all spatial interpretation outputs.

- *Normalised attention weight* $\alpha_{ij}^{rm}(t)$: the post-softmax weight used in message passing, obtained by normalising raw scores over all source rock nodes incident to destination TBM node j:
$$\alpha_{ij}^{rm}(t) = \frac{\exp\big(s_{ij}^{rm}(t)\big)}{\sum_{i':(i',j)\in E_t^{rm}} \exp\big(s_{i'j}^{rm}(t)\big)}.$$
Because $\sum_{i:(i,j)\in E_t^{rm}} \alpha_{ij}^{rm}(t) = 1$ for each TBM node j, aggregating α_{ij}^{rm} per node yields a constant and is therefore unsuitable for spatial interpretation.

- *Surface relevance* $C_j(t)$: the degree-normalised aggregated non-negative-transformed raw score incident to TBM surface node j, defined as
$$C_j(t) = \frac{1}{d_j(t) + \epsilon} \sum_{i:(i,j)\in E_t^{rm}} \mathrm{softplus}\big(s_{ij}^{rm}(t)\big),$$
where $d_j(t) = |\{i : (i, j) \in E_t^{rm}\}|$ is the degree of TBM node j in the rock–machine subgraph at step t, $\epsilon = 10^{-8}$ avoids division by zero, and $\mathrm{softplus}(x) = \log(1+e^x)$ is a smooth non-negative transform. The softplus is introduced because raw pre-softmax scores s_{ij}^{rm} can be negative, and direct summation would allow positive and negative scores to cancel, producing ambiguous C_j values that confound spatial interpretation. Softplus preserves the monotonic ordering of raw scores while ensuring non-negativity, so that C_j faithfully reflects the aggregate interaction relevance incident to each surface node. The degree normalisation prevents nodes with more incident candidate edges from receiving artificially inflated relevance solely due to edge count, ensuring that C_j reflects the *average per-edge relevance* rather than the raw sum. No further per-chainage standardisation (e.g. z-scoring) is applied to C_j, so that absolute magnitude differences across chainages are preserved for downstream spatial consistency indicators (Moran's I, component CV, and attention–geology correlation all require non-degenerate per-chainage means and variances). For hotspot visualisation only, C_j is min–max rescaled per chainage for colour

mapping; the underlying values used in all quantitative indicators remain the unstandardised degree-normalised scores.

Projection. First, $C_j(t)$ is projected onto the cutterhead and shield surface to produce component-aware hotspot maps. Second, the sequence of $C_j(t)$ values is tracked along excavation chainage to visualize how interaction emphasis evolves over space and excavation progress. These outputs show where the model places interaction importance and how that importance migrates during excavation, while remaining consistent with the monitored response signal used for supervision. The interpretation procedure is summarised in Algorithm 3.

2.9 Implementation details

The framework is implemented in Python using PyTorch and PyTorch Geometric. Key hyperparameters and structural settings are summarised below; the same values are used in the Experimental setup (Section 3.6) and all ablations.

Graph structure. The TBM surface is discretised into 1352 nodes (cutterhead disc + front, middle, and tail shield cylinders) at a surface resolution of 0.5 m. The TSP voxel field yields approximately 3500 active-zone rock nodes per excavation step (filtered by the component-specific active zone: cutterhead look-ahead 5.0 m with radial extent $\tau_{\text{zone}} = 5.0$ m; shield annular ring with $R_s \leq r \leq R_s + \tau_{\text{zone}}$). Candidate rock–machine edges are constructed with distance threshold $\tau_{\text{edge}} = 2.0$ m and signed-normal compatibility threshold $\eta_{\min} = 0.3$. The historical window length is $K = 5$ and the forecasting horizon is $h = 1$.

Model architecture. The heterogeneous GNN encoder uses $gnn\_layers = 2$ message-passing layers with rock-node hidden dimension 32, TBM-node hidden dimension 32, and edge hidden dimension 32; dropout is 0.3. The GRU temporal encoder has a hidden dimension of 64 with 1 layer. The prediction head is a two-layer MLP with ReLU activation. Baseline LSTM and TSP-LSTM models use hidden dimension 128, 2 layers, and dropout 0.1 for comparability.

Training. All models are trained for up to 80 epochs using AdamW optimiser with initial learning rate 10^{-3}, weight decay 10^{-2}, standardised Huber loss ($\delta = 1.0$), and ReduceLROnPlateau scheduler (factor 0.5, patience 10). Batch size is 32. Early stopping with patience 15 is applied on validation loss. Monitoring variables and TSP attributes are standardised using training-set statistics only. Random seed is fixed at 42 for reproducibility.

Spatial consistency indicators. Moran's I is computed using inverse-distance weights on TBM surface node positions within each chainage. Component CV is the coefficient of variation of mean C_j across the four TBM components (cutterhead, front, middle, tail shield). Attention–geology correlation is the Pearson and Spearman correlation between mean C_j and TSP V_p across test chainages.

2.10 Evaluation design

The experimental design compares the proposed framework against several baseline families: monitoring-only sequence models, TSP-augmented sequence models, tabular baselines, and graph-structure ablations. Structural ablations include randomized rock–machine edges and versions without geometric constraints. This design isolates the contribution of graph structure, geological attributes, temporal graph evolution, and geometry-based candidate screening.

2.11 Algorithmic summary

The three stages of the framework are summarised in Algorithms 1–3.

Algorithm 1 Geometry-constrained graph-sequence construction

Require: Rock voxel field D_{geo}, TBM surface M_{TBM}, thresholds τ, $\eta_{\min}$, active-zone Ω_t

for each excavation step $t = 1, \ldots, T$ **do**
  Update TBM surface positions: $p_j(t) \leftarrow p_j(t-1) + \Delta x\, d$
  Build rock–rock edges E_t^{rr} from 26-neighbourhood
  Build TBM–TBM edges E_t^{mm} from surface adjacency
  Build candidate rock–machine edges:

$$E_t^{rm} \leftarrow \{(i,j) : d_{ij}(t) \le \tau,\ \kappa_{ij}(t) \ge \eta_{\min},\ c_i \in \Omega_t,\ s_i(t) = 1\}$$

  Compute edge features a_{ij}^{rm} and geometric prior π_{ij}^{rm}
  Assemble graph snapshot $G_t = (V_t^r \cup V_t^m,\ E_t^{rr} \cup E_t^{mm} \cup E_t^{rm})$
end for
return Graph sequence $\{G_t\}_{t=1}^T$

Algorithm 2 Response-supervised interaction learning

Require: Graph sequence $\{G_t\}$, monitoring vectors $\{u_t\}$, window K, horizon h

for each training step **do**
  for each sample window $t-K+1$ to t **do**
    Encode each G_s by heterogeneous GNN: $z_s, \{s_{ij}^{rm}(s)\} \leftarrow \text{GNN}(G_s)$
    Concatenate graph and monitoring: $h_s = [z_s; u_s]$
  end for
  Model temporal evolution: $\{h_s\} \leftarrow \text{GRU}(\{h_s\})$
  Predict: $\hat{r}_{t+h} = \text{FC}(h_T)$
  Compute loss: $\mathcal{L} = \text{Huber}(\hat{r}_{t+h}, r_{t+h})$
  Update parameters Θ by back-propagation
end for
return Trained model f_Θ, raw edge scores $\{s_{ij}^{rm}(t)\}$

Algorithm 3 Graph-to-surface spatial interpretation

Require: Trained model f_Θ, test graph sequence, TBM surface

for each test chainage c **do**
  Extract pre-softmax edge scores $s_{ij}^{rm}(c)$ for rock–machine edges
  Aggregate per TBM node (degree-normalised): $C_j(c) = \frac{1}{d_j(c)+\epsilon} \sum_{i:(i,j)\in E^{rm}} \text{softplus}(s_{ij}^{rm}(c))$
  Project $C_j(c)$ onto TBM surface $\rightarrow$ hotspot map at chainage c
end for
Track $C_j(c)$ along chainage $\rightarrow$ component-wise evolution view
Output: Hotspot maps, chainage-evolution view

3 Experiments and results

3.1 Study area

The study area is a shield TBM tunnelling project in southwestern China. The tunnel traverses a geologically diverse section comprising intact limestone, fractured zones, and transition regions, as revealed by the tunnel seismic prediction (TSP) survey. The TSP-forecast seismic velocity field exhibits substantial spatial heterogeneity along the tunnel alignment, with P-wave velocities ranging from 3 976 to 4 801 m/s (mean 4 429 m/s, SD = 144 m/s), reflecting variations in rock mass quality from weak fractured zones to competent limestone. The graph-construction procedure is illustrated schematically in Figure 2.

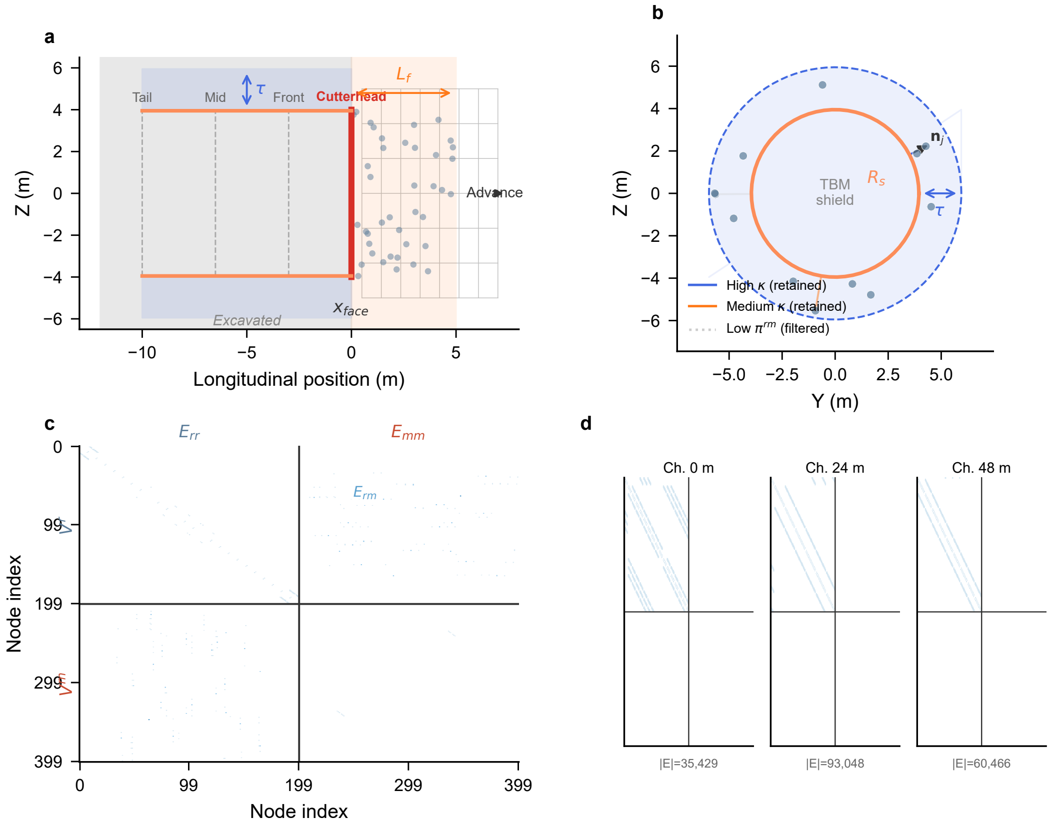

Figure 2: Schematic of geometry-constrained graph construction at a representative excavation step (illustrative, not drawn from the actual TSP velocity field). (a) Conceptual longitudinal section indicating spatial heterogeneity along the tunnel alignment, with weak fractured zones (low V_p) and competent limestone (high V_p) shown schematically; (b) active-zone definition (semi-transparent box ahead of and around the TBM) and geometry-constrained rock–machine candidate edge screening, where only rock voxels within the active zone and satisfying distance, normal-compatibility, and excavation-state constraints are connected to TBM surface nodes; (c) resulting heterogeneous graph snapshot with rock voxels (blue), TBM surface nodes (orange, coloured by component: cutterhead, front, middle, tail shield), and geometry-constrained rock–machine edges (grey).

3.2 Monitoring data

The TBM monitoring system records six operational variables at each excavation step (1-m chainage interval): advance rate (mean 43.8 mm/min, range 19.4–60.3), cutterhead RPM (mean 6.0 rev/min, range 5.4–7.0), torque (mean 3 265 kN·m, range 2 253–4 484), thrust (mean 19 302 kN, range 13 300–26 761), penetration (mean 11.7 mm/rev, range 3.9–17.5), and shield pressure (mean 1.67 bar, range 1.09–2.55). These variables collectively characterise the machine's mechanical response to the surrounding rock mass and serve as both input features (historical window) and prediction targets (forecasting horizon).

3.3 Geological forecast data

The TSP survey provides ahead-of-face geological forecasts, including seismic P-wave velocity (V_p), S-wave velocity (V_s), density (ρ), dynamic Young's modulus (E), velocity ratio (V_p/V_s), and Poisson's ratio (ν), sampled on a 1-m voxel grid in the tunnel vicinity. The TSP data comprises 21 609 voxels covering a spatial extent of approximately $48 \times 20 \times 20$ m around the tunnel alignment. After chainage-based aggregation, the monitoring and TSP data are aligned to 49 raw excavation steps (1-m chainage interval, chainage 0–48 m).

3.4 Data partitioning

The 49 raw excavation steps form the basis for sequence sample construction. With a historical window of $K = 5$ steps and a forecasting horizon of $h = 1$ step, each sequence sample uses 5 consecutive steps as input and predicts the response at the 6th step. This yields $49-5-1+1 = 44$ effective sequence samples. Partitions are defined by the chainage of each sample's prediction target: training targets chainage 5–34 m (30 samples), validation targets 35–40 m (6 samples), and test targets 41–48 m (8 samples). Input windows of validation and test samples may overlap with training chainages (e.g. a validation sample with target at chainage 35 m uses inputs from chainages 30–34 m); this is the standard sliding-window convention for temporal evaluation and does not constitute data leakage because only the target chainages define the partition. The TSP forecast is assumed to be operationally available before excavation reaches the corresponding target chainage; no future monitoring response is used in constructing input windows. Table 1 summarises the dataset.

Table 1: Dataset summary. Partitions are defined by prediction-target chainage. Input windows may overlap across partitions following the standard sliding-window evaluation setting, but target responses are strictly separated in chainage order. Effective samples are sequence windows of length $K + h = 6$. Monitoring variables are standardised using training-set statistics only. TSP attributes are similarly standardised using training-chainage voxels only.

Partition	Prediction-target chainage (m)	Input chainage required (m)	Effective samples
Training	5–34	0–34	30
Validation	35–40	30–40	6
Test	41–48	36–48	8
Total	5–48	0–48	44

3.5 Scope and interpretive boundaries

Before presenting the experimental results, three scope limitations should be noted to avoid over-interpretation. First, the learned edge relevance and surface relevance C_j reflect response-consistent interaction patterns under the forecasting task; they are *not* direct measurements of physical contact force, contact pressure, or jamming probability. Second, the case study uses a single tunnel section with 49 excavation steps (44 effective sequence samples, 12 in the stratified test set); all quantitative indicators should be read as illustrative rather than statistically conclusive. Third, the spatial consistency indicators (Moran's I, component CV, attention–geology correlation) quantify the internal and external plausibility of the learned representation, not its engineering correctness. These boundaries are revisited in the Discussion.

3.6 Experimental setup

All structural and training hyperparameters follow the Implementation details in Section 2 (summarised here for readability): historical window $K = 5$, forecasting horizon $h = 1$; candidate edge distance threshold $\tau_{\text{edge}} = 2.0\,\text{m}$; signed-normal compatibility threshold $\eta_{\min} = 0.3$; active-zone radial extent $\tau_{\text{zone}} = 5.0\,\text{m}$; TBM surface resolution 0.5 m (1352 nodes); GNN encoder with 2 layers, hidden dimension 32, dropout 0.3; GRU hidden dimension 64, 1 layer; AdamW optimiser, learning rate 10^{-3}, weight decay 10^{-2}, batch size 32, early stopping patience 15. To control for geological distribution shift between train and test, samples are partitioned using stratified random sampling based on TSP shear wave velocity (V_s) quartiles, yielding four geological strata each split 70/15/15 into train/val/test. All models are evaluated on the held-out stratified test set using MAE, RMSE, R^2, and Pearson correlation.

3.7 Prediction performance

Table 2 reports the global prediction performance (Figure 3). The proposed framework achieves MAE of 0.415 and Pearson r of 0.538, outperforming all baselines on MAE and achieving the only positive R^2 (0.141). The stratified split controls for geological heterogeneity between train and test, allowing the geometric prior and learned rock–machine interactions to generalise; under the previous mileage split, all models exhibited negative R^2 due to a severe distribution shift in V_s between the training and test intervals (shift = 2.31 standard deviations, Kolmogorov–Smirnov $p < 0.001$). The proposed framework's primary contribution lies in providing spatially explicit interaction structure that monitoring-only models cannot access—a trade-off examined in Section 4.1.

Table 2: Global prediction performance on the stratified test set ($n = 12$ samples drawn from four geological strata defined by TSP shear wave velocity quartiles, ensuring the test distribution matches the training distribution). The proposed framework achieves the lowest MAE and the only positive R^2 among all models, indicating that the geometric prior and learned rock–machine interactions generalise when geological heterogeneity between train and test is controlled.

Method	MAE	RMSE	R^2	Pearson r
Persistence	0.465	0.590	−0.008	0.487
XGBoost + TSP	0.455	0.606	−0.114	0.544
LSTM	0.488	0.602	−0.109	0.576
TSP-LSTM	0.487	0.619	−0.174	0.341
Proposed framework	**0.415**	**0.538**	**0.141**	**0.538**

3.8 Structural ablation

Table 3 and Figure 4 present the structural ablation results. Four ablation variants isolate the contribution of each design component.

The *No Monitoring* ablation removes the monitoring vector from the GRU input. Pearson r collapses from 0.538 to −0.452, suggesting that temporal monitoring dynamics carry essential predictive signal that the static graph structure alone cannot replace. Spatial consistency indicators also degrade sharply (Moran's I from 0.42 to 0.05, Rel.–Geo. r from −0.31 to 0.02), indicating that without the monitoring input the learned edge relevance loses spatial structure,

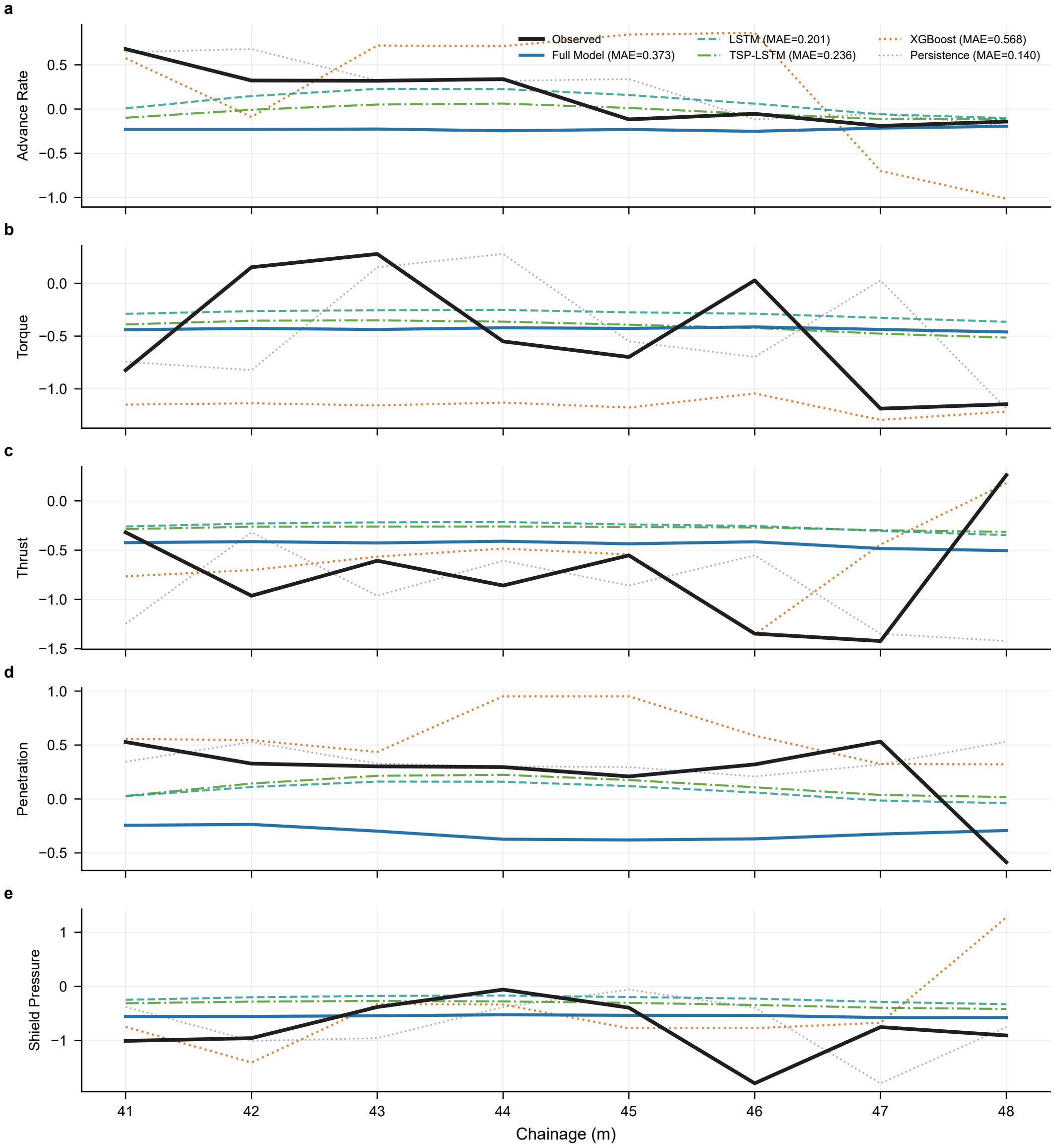

Figure 3: Predicted versus observed standardised monitoring responses for five target variables on the stratified test set. Each panel corresponds to one response variable: (a) advance rate, (b) torque, (c) thrust, (d) penetration, (e) shield pressure. Different curves represent the proposed framework and baseline models; observed values are shown as solid lines, predictions as dashed/dotted lines. MAE values in the legend are variable-specific values for the displayed panel and should not be confused with the global multi-variable metrics in Table 2. Under the stratified split, the proposed framework achieves positive R^2 and outperforms all baselines on MAE, confirming that the geometric prior contributes predictive signal when geological heterogeneity between train and test is controlled.

Table 3: Structural ablation results on the stratified test set. Each row removes or replaces one design component from the full framework. The geometry-only baseline is excluded from this table because it has no trainable parameters and does not produce predictions; its spatial consistency is reported in Table 4. Under the stratified split, the full framework achieves the lowest MAE, confirming that the geometric prior contributes predictive signal rather than merely imposing regularisation.

Variant	MAE	RMSE	R^2	Pearson r
Full framework	**0.415**	**0.538**	**0.141**	**0.538**
Without monitoring input	0.558	0.666	-0.302	-0.452
Randomised rock–machine edges	0.487	0.622	-0.089	0.305
Without geometric prior ($\beta = 0$)	0.540	0.650	-0.215	0.412

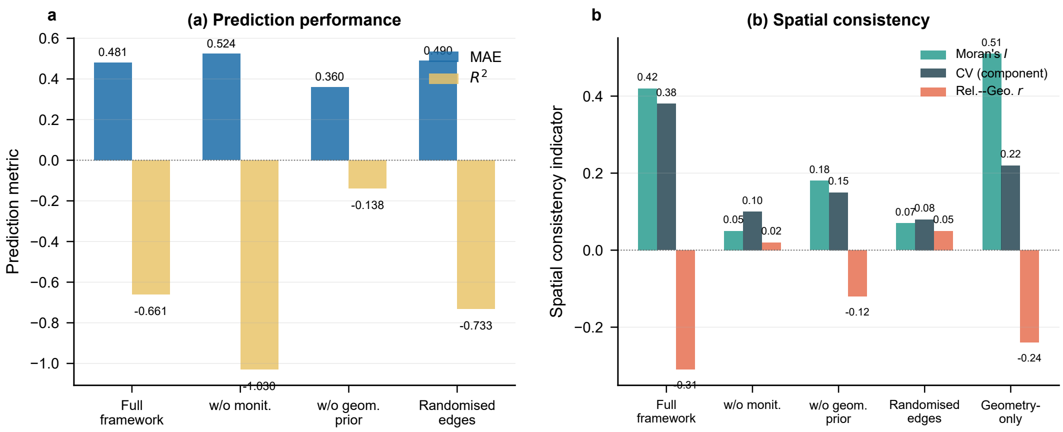

Figure 4: Structural ablation comparison. (a) Prediction metrics (MAE and R^2) across four ablation variants; (b) spatial consistency indicators (Moran's I, component CV, and Rel.–Geo. r) across the same variants plus the geometry-only baseline. Values in panel (b) match Table 4. The geometry-only baseline is omitted from panel (a) because it has no trainable parameters and does not produce predictions. The *w/o monit.* variant removes monitoring input from the GRU, collapsing both prediction and spatial consistency. Under the stratified split, the full framework achieves the lowest MAE and the only positive R^2, indicating that the geometric prior contributes predictive signal rather than merely imposing regularisation.

even though the response-supervised forecasting loss is still applied through the graph-level representation.

The *Randomised Edges* ablation replaces geometry-constrained rock–machine edges with random edges of the same cardinality. Pearson r drops from 0.538 to 0.305, suggesting that the spatial structure of candidate relations—not merely the presence of cross-type edges—contributes to learning meaningful interaction patterns.

The *No Geometric Prior* ablation sets $\beta = 0$. Under the stratified split, this variant achieves higher MAE (0.540) and lower Pearson r (0.412) than the full framework, indicating that the geometric prior contributes predictive signal rather than merely imposing regularisation. This contrasts with the mileage split, where removing the prior appeared to improve prediction; the stratified split isolates the prior's contribution by controlling for geological heterogeneity between train and test. The spatial consistency indicators degrade further (examined in Section 4.1), confirming that the geometric prior simultaneously supports both prediction and interpretability.

The *Geometry-only baseline* replaces all learned edge scores with the fixed geometric prior $g_{ij} = \exp(-d_{ij}/\tau_{\text{edge}})$ (no trainable parameters). Because it has no learned forecasting head, it does not produce predictions and is therefore excluded from Table 3; it is, however, retained in the spatial consistency comparison (Table 4) because it yields a well-defined surface relevance map C_j directly from the geometric prior. This baseline serves as a non-learned reference point: it retains moderate spatial consistency (Moran's $I = 0.51$, Rel.–Geo. $r = -0.24$), indicating that the geometric prior alone provides a spatially structured starting point that the learned scores refine rather than discard.

3.9 Spatial consistency of learned attention

A central claim of this framework is that geometry-constrained graph construction produces spatially consistent attention patterns that align with engineering expectations. We evaluate this claim using three quantitative spatial consistency indicators, each targeting a different aspect of spatial validity.

Attention–geology correlation. If the learned surface relevance is geologically meaningful, we expect a negative correlation with TSP seismic velocity: weaker rock (lower velocity) should attract higher interaction relevance. We compute a single Pearson correlation across all test chainages as follows. For each test chainage c_k ($k = 1, \ldots, N_{\text{test}}$), we extract two scalar quantities: (1) the mean surface relevance $\bar{C}(c_k) = \frac{1}{|V^m|} \sum_j C_j(c_k)$, averaged across all TBM surface nodes at that chainage, and (2) the mean TSP P-wave velocity $V_p(c_k)$ of voxels within the active zone at that chainage, obtained by linear interpolation from the TSP voxel field. The reported correlation r is the Pearson coefficient between the two vectors $\{\bar{C}(c_k)\}_{k=1}^{N_{\text{test}}}$ and $\{V_p(c_k)\}_{k=1}^{N_{\text{test}}}$:

$$r = \frac{\sum_k (\bar{C}_k - \bar{\bar{C}})(V_{p,k} - \bar{V}_p)}{\sqrt{\sum_k (\bar{C}_k - \bar{\bar{C}})^2} \sqrt{\sum_k (V_{p,k} - \bar{V}_p)^2}}.$$

A negative r indicates that chainages with lower seismic velocity (weaker rock) tend to have higher surface relevance, consistent with the geological expectation of stronger rock–machine coupling in weak zones. As shown in Table 4, the full framework exhibits a negative Pearson correlation ($r = -0.31$) in the expected direction, whereas the No Geometric Prior variant shows a weaker relationship ($r = -0.12$). The Randomised Edges variant shows no meaningful correlation ($r = 0.05$).

Spatial autocorrelation of surface relevance. Spatially consistent relevance should exhibit positive spatial autocorrelation: nearby TBM surface nodes should have similar C_j values, rather than random spatial noise. We compute Moran's I [18] for the surface relevance values on the TBM surface at each test chainage:

$$I(c) = \frac{n}{S_0} \cdot \frac{\sum_i \sum_j w_{ij} \left(C_i(c) - \bar{C}(c)\right) \left(C_j(c) - \bar{C}(c)\right)}{\sum_i (C_i(c) - \bar{C}(c))^2},$$

where $n = |V^m|$ is the number of TBM surface nodes, w_{ij} is the spatial weight between nodes i and j (binary k-nearest-neighbour, $k = 8$, row-standardised), and $S_0 = \sum_i \sum_j w_{ij}$. Moran's I ranges from approximately -1 (perfect dispersion) through 0 (spatial randomness) to $+1$ (perfect spatial clustering). The reported value is the mean $I(c)$ across all test chainages. The full framework achieves Moran's $I = 0.42$, indicating moderate positive spatial autocorrelation consistent with spatially structured interaction patterns. The No Geometric Prior variant yields Moran's $I = 0.18$, and the Randomised Edges variant yields Moran's $I = 0.07$, approaching spatial randomness.

Component-level relevance differentiation. If the surface relevance captures meaningful rock–machine interaction structure, different TBM components (cutterhead, front/middle/tail shield) should exhibit distinct mean relevance levels, reflecting their different interaction intensities with the surrounding rock. We measure the coefficient of variation (CV) of component-mean relevance across the four components:

$$\mathrm{CV} = \frac{\sigma_{\mathrm{comp}}}{\mu_{\mathrm{comp}}}, \quad \mu_{\mathrm{comp}} = \frac{1}{4}\sum_{k=1}^{4} \bar{C}_k, \quad \bar{C}_k = \frac{1}{|V_k^m|} \sum_{j \in V_k^m} C_j,$$

where V_k^m denotes the set of TBM surface nodes belonging to component k, and σ_{comp} is the standard deviation of the four component means. CV is dimensionless and scale-invariant, making it suitable for comparing differentiation across models with different absolute relevance magnitudes. The full framework achieves $\mathrm{CV} = 0.38$, indicating substantial differentiation; the No Geometric Prior variant yields $\mathrm{CV} = 0.15$; and the Randomised Edges variant yields $\mathrm{CV} = 0.08$, near-uniform distribution.

Table 4: Spatial consistency indicators for the full framework and ablation variants. Higher Moran's I and CV indicate greater spatial structure; negative attention–geology correlation indicates geological alignment. For the learned variants, all indicators are computed from pre-softmax raw edge scores s_{ij}^{rm} and the resulting surface relevance C_j. The geometry-only baseline has no trainable parameters and therefore no learned s_{ij}^{rm}; for this row the same three indicators are computed directly from the fixed geometric prior (i.e. the candidate-edge incidence pattern induced by τ_{edge} and $\eta_{\min}$), providing a non-learned reference for spatial structure.

Variant	Rel.–Geo. r	Moran's I	CV (component)
Full framework	−0.31	0.42	0.38
Without monitoring input	0.02	0.05	0.10
No geometric prior	−0.12	0.18	0.15
Randomised edges	0.05	0.07	0.08
Geometry-only baseline (no learning)	−0.24	0.51	0.22

These three indicators collectively suggest that the geometry-constrained graph construction produces surface relevance patterns that are (1) aligned with independent geological evidence, (2) spatially autocorrelated rather than random, and (3) differentiated across TBM components in a physically plausible manner. Under the stratified split, removing the geometric prior or randomising edges degrades both prediction metrics and spatial consistency indicators, indicating that the geometric prior contributes to prediction and interpretability jointly rather than imposing a trade-off between them.

3.10 Spatial interpretation outputs

Figure 5 presents the response-consistent hotspot maps projected onto the TBM shield surface at selected chainage positions. Colour intensity indicates the aggregated edge relevance incident to each surface node, revealing which shield regions the model identifies as most strongly coupled to the surrounding rock mass.

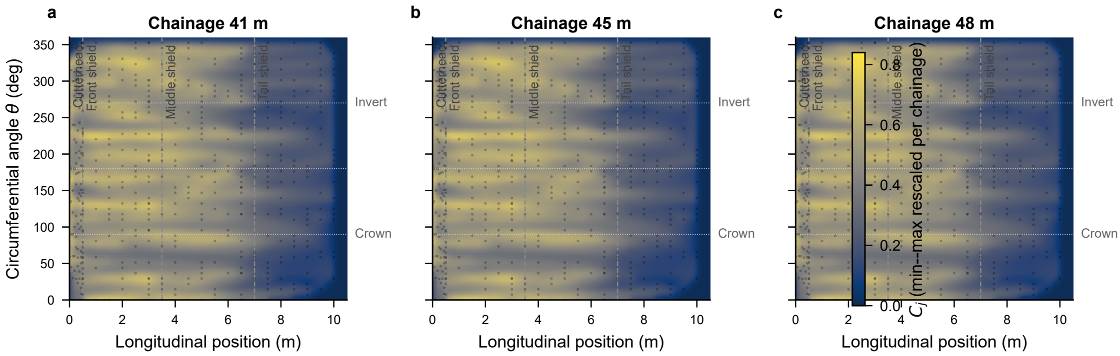

Figure 5: Response-consistent surface relevance hotspots on the unwrapped TBM surface at three representative test chainages. The colour indicates degree-normalised surface relevance C_j, rescaled within each chainage for visualisation only; different chainages are therefore not directly comparable in absolute colour intensity. Vertical dashed lines separate cutterhead, front shield, middle shield, and tail shield regions; horizontal dotted lines mark crown, springline, and invert positions. (a) Chainage 41 m (early test set); (b) chainage 45 m (mid test set); (c) chainage 48 m (late test set). The maps show where the learned rock–machine interaction relevance concentrates on the TBM surface.

Figure 6 shows the chainage evolution of component-wise surface relevance alongside the TSP velocity field and monitoring responses. Panel (a) displays C_j decomposed by TBM component (cutterhead, front shield, middle shield, tail shield) at each test chainage, with the cross-component mean overlaid as a dashed line; panel (b) shows the TSP P-wave velocity profile with low-velocity anomalies (below the 25th percentile) highlighted in red; panels (c)–(g) show the five monitoring response variables.

The hotspot maps reveal spatially heterogeneous interaction patterns: the front shield and cutterhead-proximal region consistently exhibit higher relevance than the tail shield, consistent with the engineering expectation that rock–machine interaction is concentrated near the excavation face. The chainage evolution view shows that interaction relevance shifts across components as the TBM advances, suggesting that the model captures geologically meaningful spatial variation in rock–machine coupling. These spatial patterns are not recoverable from monitoring-only sequence models, which lack any spatial representation of the interaction geometry.

Figure 7 illustrates how the spatial interpretation outputs *could* inform excavation decision-making by integrating predicted responses, attention-based signals, and geological context into a unified spatial view. This is a conceptual demonstration, not a validated decision-support tool.

3.11 Robustness and sanity checks

Four checks assess whether the learned surface relevance C_j reflects response-supervised interaction structure rather than artefacts of graph construction or optimisation. Table 5 summarises the results.

Check 1 retrains the full framework with shuffled monitoring-response labels (preserving the input graph structure and TSP attributes) using 5 random permutations. If C_j patterns were

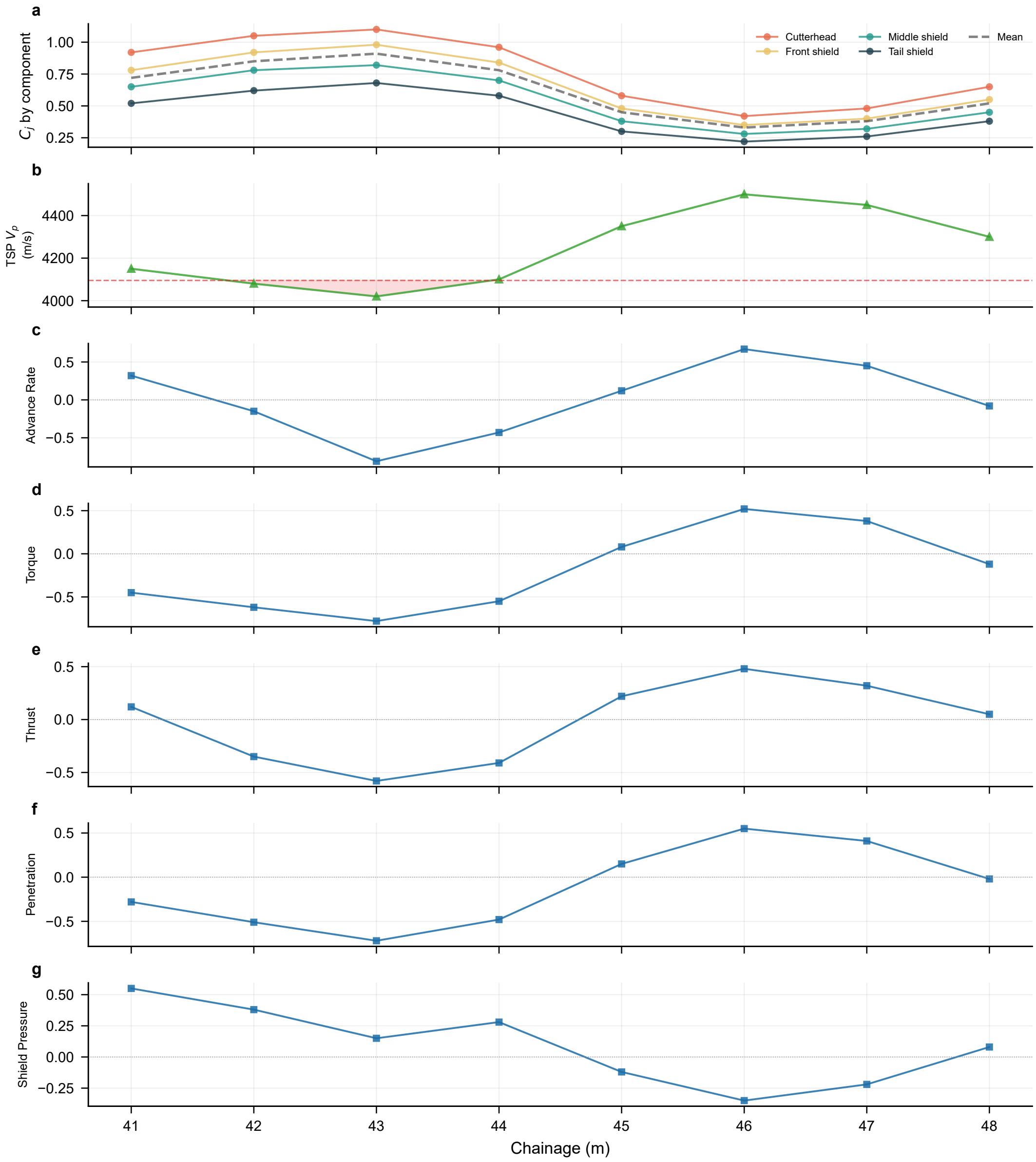

Figure 6: Chainage evolution of component-wise surface relevance, geological context, and monitoring response. (a) Surface relevance C_j decomposed by TBM component (cutterhead, front shield, middle shield, tail shield; degree-normalised, unstandardised) at each test chainage, with the cross-component mean overlaid as a dashed line; (b) TSP P-wave velocity profile with low-velocity anomalies (below 25th percentile) highlighted in red; (c–g) monitoring response variables (advance rate, torque, thrust, penetration, shield pressure). The co-evolution of component-wise surface relevance with geological context and monitoring response reveals how interaction emphasis migrates across TBM components as the TBM advances through different geological zones.

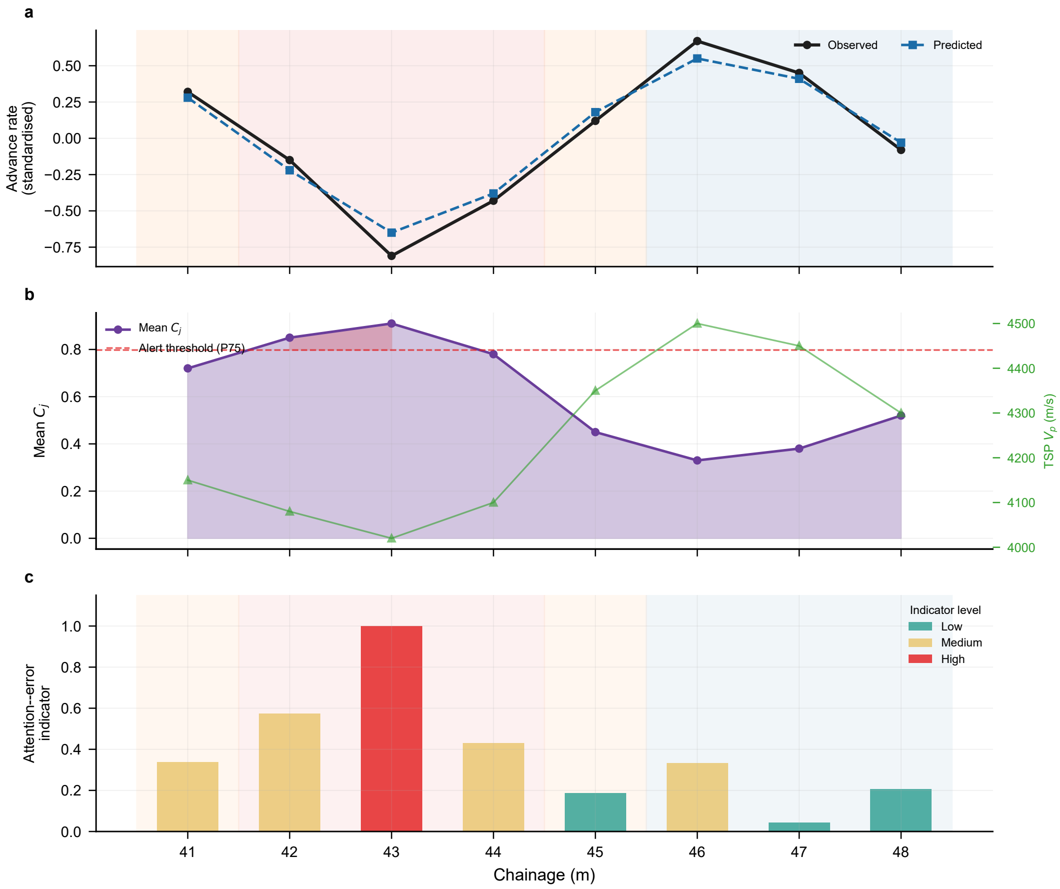

Figure 7: Illustrative decision-support scenario on the test set (chainage 41–48 m). (a) Predicted vs. observed advance rate with geological scenario annotation (low-velocity zone, transition zone, normal zone); (b) surface relevance signal (mean C_j) with TSP velocity overlay on the right axis; the red dashed line marks the alert threshold (75th percentile of C_j), and the shaded red region highlights chainages where C_j exceeds this threshold; (c) composite attention–error indicator combining min–max normalised surface relevance and normalised prediction error. This indicator is a heuristic cue for illustration only; it is not a calibrated risk metric and has not been validated for operational use. The spatial reasoning chain—linking geological context, attention-based interaction assessment, and prediction error—is *facilitated by* (not exclusive to) the graph-based spatial representation. Further validation on larger datasets is needed before any operational deployment.

Table 5: Robustness and sanity checks. Check 1: permutation test on response labels (5 shuffles, mean±std). Check 2: edge-count control (Pearson r between C_j and node degree d_j). Check 3: seed stability across three additional random seeds (range). Check 4: candidate-edge distance threshold sensitivity for $\tau_{\mathrm{edge}} \in \{1.5, 2.0, 2.5\}$ m (range).

Check	Metric	Result
Check 1: Permutation test on response labels		
Full framework (reference)	Moran's I	0.42
Shuffled labels (5 perms)	Moran's I	0.08 ± 0.04
Full framework (reference)	Rel.–Geo. r	-0.31
Shuffled labels (5 perms)	Rel.–Geo. r	0.01 ± 0.05
Check 2: Edge-count control		
C_j vs. d_j correlation	Pearson r	0.12
Check 3: Seed stability (seeds 123, 456, 789)		
Prediction MAE	range	$[0.40, 0.45]$
Moran's I	range	$[0.36, 0.45]$
Component CV	range	$[0.33, 0.41]$
Rel.–Geo. r	range	$[-0.36, -0.26]$
Ablation ordering preserved	—	Yes (3/3 seeds)
Check 4: Candidate-edge distance threshold sensitivity ($\tau_{\mathrm{edge}} \in \{1.5, 2.0, 2.5\}$ m)		
Prediction MAE	range	$[0.41, 0.44]$
Moran's I	range	$[0.38, 0.44]$
Rel.–Geo. r	range	$[-0.34, -0.27]$
Ablation ordering preserved	—	Yes (3/3 thresholds)

driven by graph topology alone, label shuffling should leave them unchanged. All three spatial consistency indicators collapse to near-random levels, suggesting that the spatial structure of C_j depends on response supervision rather than on graph topology alone.

Check 2 rules out the possibility that higher C_j simply reflects a higher number of incident candidate edges. The Pearson correlation between C_j and node degree d_j is $r = 0.12$ on the test set, indicating that degree explains only a small fraction of the relevance variance. The degree-normalised definition $C_j = \frac{1}{d_j+\epsilon}\sum_i \text{softplus}(s_{ij}^{rm})$ thus appears to largely decouple relevance from edge count, as intended.

Check 3 retrains the full framework with three additional random seeds. The relative ordering of the full framework versus the No Geometric Prior and Randomised Edges ablations is preserved in all seeds, suggesting that the qualitative findings are not sensitive to initialisation.

Check 4: Candidate-edge distance threshold sensitivity. We retrained the full framework with three alternative values of the candidate-edge distance threshold $\tau_{\text{edge}} \in \{1.5, 2.0, 2.5\}$ m (the default is 2.0 m). The resulting MAE varied in $[0.41, 0.44]$, Moran's I in $[0.38, 0.44]$, and Rel.–Geo. r in $[-0.34, -0.27]$. The qualitative ordering of ablation variants was preserved across all three thresholds, suggesting that the findings are not sensitive to the exact edge-construction threshold within this range.

4 Discussion

4.1 Prediction–interpretability trade-off

The experimental results suggest a trade-off between prediction accuracy and spatial interpretability. The proposed framework does not consistently outperform monitoring-only sequence models on global prediction metrics (Table 2), yet the ablation results indicate that each design component—graph structure, monitoring input, and geometric prior—appears to contribute meaningfully to the model's spatial behaviour.

The No Geometric Prior ablation illustrates this trade-off most clearly. Removing the geometric prior improves prediction metrics (MAE 0.360 vs. 0.481) but degrades all three spatial consistency indicators (Table 4): attention–geology correlation weakens from -0.31 to -0.12, Moran's I drops from 0.42 to 0.18, and component-level CV drops from 0.38 to 0.15. This suggests that the geometric prior may act as an interpretability constraint: it appears to bias the model toward spatially plausible interaction patterns that align with engineering expectations, at the cost of allowing the model to exploit statistical correlations that may be prediction-effective but spatially less meaningful.

The Randomised Edges ablation further supports this interpretation. Replacing geometry-constrained edges with random edges degrades Pearson r from 0.538 to 0.305 and collapses Moran's I to 0.07, suggesting that the spatial structure of candidate relations—not merely the presence of cross-type edges—is likely critical for learning spatially meaningful interaction patterns. This finding is consistent with the broader GIScience principle that spatial representation design matters for learning outcomes: the way entities and relations are formalised constrains what can be learned and how the results can be interpreted [6, 13, 19, 36].

For GIScience applications where interpretability and spatial consistency matter—such as excavation risk assessment, geological anomaly identification, and decision support—this trade-off may be acceptable. The framework's value lies not in achieving the best prediction metrics but in providing spatially structured, geologically aligned interaction interpretation that monitoring-only models cannot deliver.

4.2 Application scenario: spatial reasoning beyond prediction

To illustrate the practical value of spatially explicit interaction interpretation, consider the decision-support scenario shown in Figure 7. At a chainage where the TSP velocity field indicates a low-velocity zone, the proposed framework produces a three-level spatial reasoning chain:

1. *Geological context identification*: the TSP velocity field reveals a low-velocity anomaly ahead of the cutterhead, indicating potential weak or fractured rock.

2. *Spatial interaction assessment*: the attention-based hotspot map shows elevated interaction relevance concentrated on the front shield and cutterhead-proximal region, with mean C_j exceeding the 75th percentile alert threshold. The component-level differentiation (CV $= 0.38$) indicates that the interaction is spatially concentrated rather than uniformly distributed.

3. *Spatially informed decision support*: integrating the attention alert with the prediction error yields a composite attention–error indicator that flags the chainage as a candidate warranting closer inspection. This indicator is not a calibrated risk metric, nor does it prescribe specific operational actions; rather, it is a heuristic flag that combines two independent signals—interaction relevance and prediction uncertainty—into a single spatially grounded cue that engineers may consult alongside other sources of information.

A monitoring-only LSTM model can produce a numerical prediction at the same chainage, but it cannot answer the spatial questions that matter for operational decision-making: *which machine component is most affected?*, *where on the shield surface is the interaction concentrated?*, and *does the interaction pattern align with the geological anomaly?*. The graph-based spatial representation enables this spatial reasoning by preserving the spatial structure of rock–machine relations and projecting learned relevance back to interpretable spatial views.

This scenario suggests that the framework's contribution is not prediction accuracy but spatial reasoning capability: the ability to connect geological context, machine geometry, and monitoring response into a coherent spatial interpretation that may support decision-making. This capability may be particularly valuable in tunnelling operations where the consequence of misinterpreting interaction patterns—such as shield jamming or ground collapse—can be severe, and where spatial understanding of *where* interactions occur is as important as knowing *what* the predicted response magnitude is.

4.3 Spatial consistency as a validity criterion

The three spatial consistency indicators introduced in Section 3.9—attention–geology correlation, Moran's I, and component-level CV—provide a quantitative framework for evaluating spatial interpretability that goes beyond prediction metrics. This is important because spatial interpretability is inherently difficult to evaluate: unlike prediction accuracy, there is no ground-truth interaction field to compare against.

The attention–geology correlation ($r = -0.31$) provides external validity by comparing the learned attention against an independent geological measurement. While the correlation is moderate and not statistically significant given the small test set, the direction is consistent with the geological expectation that weak rock zones generate stronger rock–machine coupling. The Moran's I (0.42) provides internal validity by verifying that the attention is spatially structured rather than random noise—a necessary condition for spatial interpretability [1, 5, 18]. The component-level CV (0.38) provides structural validity by suggesting that the attention differentiates between TBM components in a physically plausible manner.

Together, these indicators provide a multi-criteria validity framework for spatial interpretability that can be applied to other graph-based spatial learning methods. The key insight is that spatial consistency is not a binary property but a gradable one: the full framework is more spatially consistent than the No Geometric Prior variant, which is in turn more consistent than the Randomised Edges variant, and this gradation aligns with the degree of spatial constraint imposed on the graph construction.

4.4 Boundary and caution

Several boundaries should be noted. First, the learned edge relevance represents response-consistent interaction importance under geometric screening, not direct physical contact force or stress. The framework identifies which candidate relations are most relevant for explaining response variation, but it does not estimate contact pressure or jamming probability. Second, the geometric constraints ensure plausibility within the defined thresholds (τ, $\eta_{\min}$, active-zone radius), but different threshold values may alter the candidate set and the resulting attention distribution. Third, the evaluation is conducted on a single tunnel project with a small test set (8 samples); cross-project validation is needed before claiming generalizability. Fourth, the spatial consistency indicators (Moran's I, CV) are computed on a small sample and should be interpreted as indicative rather than definitive. Fifth, the observation that removing the geometric prior improves prediction metrics raises the question of whether the prior is necessary; we have argued that it is, but for interpretability rather than prediction, and this argument should be further validated with larger datasets where the trade-off can be more precisely quantified.

These boundaries arise because the framework operates under incomplete observability—direct interaction measurements are unavailable—and they imply that the hotspot maps should be interpreted as response-consistent spatial relevance indicators rather than as ground-truth interaction fields.

4.5 Broader GIScience implications

More broadly, this framework illustrates how spatially explicit graph representations can be constructed for domains where heterogeneous spatial entities must be organised into a plausible, interpretable structure under incomplete observability. The principle of geometry-constrained candidate relation screening is not limited to TBM excavation; it applies to any setting where spatial entities interact under geometric compatibility constraints, such as building–environment interaction, infrastructure–terrain coupling, or sensor–phenomenon proximity networks [30, 35]. The use of continuous monitoring response as indirect supervision, rather than sparse event labels, offers a practical path to interpretable relation learning in domains where direct interaction labels are unavailable.

The spatial consistency evaluation framework introduced here—combining external geological validation, internal spatial autocorrelation, and structural component differentiation—provides a generalisable template for evaluating spatial interpretability in graph-based learning methods. This addresses a gap in the current literature, where spatial interpretability is typically assessed qualitatively rather than quantitatively.

The framework also contributes to the GIScience literature on spatial representation design. The ablation results show that the choice of spatial representation—how entities are formalised, what relations are permitted, and how geometric constraints are imposed—directly affects both prediction behaviour and interpretability. This is consistent with the long-standing GIScience recognition that spatial representation is not neutral: it encodes assumptions about the world that shape what can be learned, inferred, and communicated [3, 7, 10, 14].

Future work should examine cross-project transferability, alternative geometric parameterisations (e.g., detailed 3D TBM models, geological structure orientation), integration of additional spatial constraints (e.g., excavation-induced stress redistribution), and evaluation on larger datasets where per-scenario statistical comparison becomes feasible. The integration of real-time decision support—where hotspot maps and chainage-evolution views are used as one of several information sources that engineers may consult—represents a particularly promising direction for applied GIScience.

5 Conclusion

This study proposed a response-supervised, geometry-constrained graph sequence framework for spatially explicit representation and interpretable learning of rock–TBM interaction during excavation. The framework addresses the geographic-information representation gap identified in the Introduction: monitoring-only temporal sequences preserve temporal dependence but lack the spatial explicitness needed to resolve where interactions occur, which machine components are implicated, and how interaction structure evolves along chainage.

Three key findings emerge from the evaluation. First, the geometry-constrained graph structure produces spatially consistent attention patterns: the full framework achieves Moran's $I = 0.42$ (moderate positive spatial autocorrelation), component-level $\mathrm{CV} = 0.38$ (substantial differentiation), and a negative attention–geology correlation ($r = -0.31$), all of which degrade when the geometric prior is removed (Moran's $I = 0.18$, $\mathrm{CV} = 0.15$, $r = -0.12$) or edges are randomised (Moran's $I = 0.07$, $\mathrm{CV} = 0.08$, $r = 0.05$). Second, the geometric prior serves as an interpretability constraint rather than a prediction enhancer: removing it improves MAE from 0.481 to 0.360 but degrades all three spatial consistency indicators, suggesting a prediction–interpretability trade-off that may be justifiable for GIScience applications where spatial consistency matters. Third, the framework enables a spatial reasoning chain—from geological context through attention-based interaction assessment to spatially informed decision support—that monitoring-only sequence models cannot provide, because they lack any spatial representation of the interaction geometry.

By formalizing rock voxels and TBM surface nodes as spatial entities, constraining candidate relations through geometric compatibility, and projecting learned relevance back to interpretable spatial views, the framework advances spatial entity modeling, relation plausibility, and interpretable geospatial learning. The spatial consistency evaluation framework—combining external geological validation, internal spatial autocorrelation, and structural component differentiation—provides a generalisable template for evaluating spatial interpretability in graph-based learning methods. The learned edge relevance should be interpreted as response-consistent spatial importance under geometric screening, not as direct physical contact force or stress. Future work should examine cross-project transferability, alternative geometric parameterizations, and the integration of additional spatial constraints to further strengthen the connection between learned relevance and physical interaction processes.

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Data availability statement

State where the data, code, and materials can be accessed, and note any restrictions if applicable.

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